

# **ECEN-601: Homework 1**

Due on September 12, 2025 at 5:00 pm

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## Problem 1

Which of the following expressions are statements?

- (a) Today is a nice day. (**Not a statement**)
- (b) Go to sleep. (**Not a statement**)
- (c) Is it going to snow tomorrow? (**Not a statement**)
- (d) The U.S. has 49 states. (**Statement**)
- (e) I like to eat fruit, and you often think about traveling to Spain. (**Not a statement**)
- (f) If we go out tonight, the babysitter will be unhappy. (**Not a statement**)
- (g) Call me on Thursday if you are home. (**Not a statement**)

## Solution

Only **(d)** is a statement.

I feel like **(a)** *could* be a statement, though it varies from person-to-person (similar to **(e)**), so I say not a statement. Additionally, **(f)** is a conditional statement, but I don't think that's what the question is asking.

**Problem 2**

Let  $X$  = "I am happy," let  $Y$  = "I am watching a movie" and  $Z$  = "I am eating spaghetti." Translate the following statements into words.

- (a)  $Z \rightarrow X$ .
- (b)  $X \leftrightarrow Y$ .
- (c)  $(Y \vee Z) \rightarrow X$ .
- (d)  $Y \vee (Z \rightarrow X)$ .
- (e)  $(Y \rightarrow \neg X) \wedge (Z \rightarrow \neg X)$ .
- (f)  $(X \wedge \neg Y) \leftrightarrow (Y \vee Z)$ .

**Solution**

**Part (a):** If I am eating spaghetti, then I am happy.

**Part (b):** I am happy if and only if I am watching a movie.

**Part (c):** If I am watching a movie or I am eating spaghetti, then I am happy.

**Part (d):** If I am eating spaghetti then I am happy, or I am watching a movie.

**Part (e):** If I am watching a movie then I am not happy, and if I am eating spaghetti then I am not happy.

**Part (f):** I am happy and I am not watching a movie, if and only if I am watching a movie or I am eating spaghetti.

### Problem 3

Let  $E$  = "The house is blue," let  $F$  = "The house is 30 years old" and  $G$  = "The house is ugly."  
Translate the following statements into symbols.

- (a) If the house is 30 years old, then it is ugly.
- (b) If the house is blue, then it is ugly or it is 30 years old.
- (c) If the house is blue, then it is ugly, or it is 30 years old.
- (d) The house is not ugly if and only if it is 30 years old.
- (e) The house is 30 years old if it is blue, and it is not ugly if it is 30 years old.
- (f) For the house to be ugly, it is necessary and sufficient that it be ugly and 30 years old.

### Solution

**Part (a):**  $F \rightarrow G$ .

**Part (b):**  $E \rightarrow (G \vee F)$

**Part (c):**  $(E \rightarrow G) \vee F$

**Part (d):**  $\neg G \leftrightarrow F$

**Part (e):**  $(E \rightarrow F) \wedge (F \rightarrow \neg G)$

**Part (f):**  $G \leftrightarrow (G \wedge F)$

## Problem 4

Make a truth table for each of the following statements.

- (a)  $P \wedge \neg Q$ .
- (b)  $(R \vee S) \wedge \neg R$ .
- (c)  $X \vee (\neg Y \vee Z)$ .
- (d)  $(A \vee B) \wedge (A \vee C)$ .
- (e)  $(P \wedge R) \vee \neg(Q \wedge S)$ .

## Solution

I have constructed these truth tables in a manner similar to the truth tables in Engineering Fundamentals. Additionally, I have highlighted the final column of each truth table in light green.

### Part (a)

| $P$ | $Q$ | $P$ | $\wedge$ | $\neg Q$ |
|-----|-----|-----|----------|----------|
| T   | T   | T   | T        | F        |
| T   | F   | T   | F        | T        |
| F   | T   | F   | F        | F        |
| F   | F   | F   | F        | T        |

### Part (b)

| $R$ | $S$ | $(R$ | $\vee$ | $S)$ | $\wedge$ | $\neg R$ |
|-----|-----|------|--------|------|----------|----------|
| T   | T   | T    | T      | T    | F        | F        |
| T   | F   | T    | T      | F    | F        | F        |
| F   | T   | F    | T      | T    | T        | T        |
| F   | F   | F    | F      | F    | F        | T        |

### Part (c)

| $X$ | $Y$ | $Z$ | $X$ | $\vee$ | $(\neg Y$ | $\vee$ | $Z)$ |
|-----|-----|-----|-----|--------|-----------|--------|------|
| T   | T   | T   | T   | T      | F         | T      | T    |
| T   | T   | F   | T   | T      | F         | F      | F    |
| T   | F   | T   | T   | T      | T         | T      | T    |
| T   | F   | F   | T   | T      | T         | T      | F    |
| F   | T   | T   | F   | T      | F         | T      | T    |
| F   | T   | F   | F   | F      | F         | F      | F    |
| F   | F   | T   | F   | T      | T         | T      | T    |
| F   | F   | F   | F   | T      | T         | T      | F    |

### Part (d)

| $A$ | $B$ | $C$ | $(A$ | $\vee$ | $B)$ | $\wedge$ | $(A$ | $\vee$ | $C)$ |
|-----|-----|-----|------|--------|------|----------|------|--------|------|
| T   | T   | T   | T    | T      | T    | T        | T    | T      | T    |
| T   | T   | F   | T    | T      | T    | T        | T    | T      | F    |
| T   | F   | T   | T    | T      | F    | T        | T    | T      | T    |
| T   | F   | F   | T    | T      | F    | T        | T    | T      | F    |
| F   | T   | T   | F    | T      | T    | T        | F    | T      | T    |
| F   | T   | F   | F    | T      | T    | F        | F    | F      | F    |
| F   | F   | T   | F    | F      | F    | F        | F    | T      | T    |
| F   | F   | F   | F    | F      | F    | F        | F    | F      | F    |

*Final truth table on following page...*

**Part (e)**

| $P$ | $Q$ | $R$ | $S$ | $(P \wedge R)$ | $\vee$ | $\neg$ | $(Q \wedge S)$ |
|-----|-----|-----|-----|----------------|--------|--------|----------------|
| T   | T   | T   | T   | T              | T      | F      | T              |
| T   | T   | T   | F   | T              | T      | T      | F              |
| T   | T   | F   | T   | F              | F      | T      | T              |
| T   | T   | F   | F   | F              | T      | T      | F              |
| T   | F   | T   | T   | T              | T      | F      | F              |
| T   | F   | T   | F   | T              | T      | F      | F              |
| T   | F   | F   | T   | F              | T      | F      | F              |
| T   | F   | F   | F   | F              | T      | F      | F              |
| F   | T   | T   | T   | F              | F      | T      | T              |
| F   | T   | T   | F   | F              | T      | T      | F              |
| F   | T   | F   | T   | F              | F      | T      | T              |
| F   | T   | F   | F   | F              | T      | T      | F              |
| F   | F   | T   | T   | F              | T      | F      | F              |
| F   | F   | T   | F   | F              | T      | F      | F              |
| F   | F   | F   | T   | F              | T      | F      | F              |
| F   | F   | F   | F   | F              | T      | F      | F              |

## Problem 5

Simplify the following statements (making use of any equivalence of statements given so far in the text or exercises).

- (a)  $\neg(P \rightarrow \neg Q)$
- (b)  $A \rightarrow (A \wedge B)$
- (c)  $(X \wedge Y) \rightarrow X$
- (d)  $\neg(M \vee L) \wedge L$
- (e)  $(P \rightarrow Q) \vee Q$
- (f)  $\neg(X \rightarrow Y) \vee Y$

## Solution

These simplifications can be found pretty quickly by writing out the truth tables, though I will omit that process and use the stated identities in the text and notes.

**Part (a):**  $\neg(P \rightarrow \neg Q) \equiv P \wedge \neg(\neg Q) \equiv P \wedge Q.$

**Part (b):**  $A \rightarrow (A \wedge B) \equiv \neg A \vee (A \wedge B) \equiv (\neg A \vee A) \wedge (\neg A \vee B) \equiv \neg A \vee B.$

**Part (c):**  $(X \wedge Y) \rightarrow X \equiv \neg(X \wedge Y) \vee X \equiv \underbrace{(\neg X \vee X)}_{\text{tautology}} \vee (\neg Y) \equiv \text{TRUE}.$

**Part (d):**  $\neg(M \vee L) \wedge L \equiv (\neg M \wedge \neg L) \wedge L \equiv \underbrace{(\neg L \wedge L)}_{\text{contradiction}} \wedge L \equiv \text{FALSE}.$

**Part (e):**  $(P \rightarrow Q) \vee Q \equiv (\neg P \vee Q) \vee Q \equiv (Q \vee Q) \vee \neg P \equiv Q \vee \neg P$

**Part (f):**  $\neg(X \rightarrow Y) \vee Y \equiv (X \wedge \neg Y) \vee Y \equiv \underbrace{(Y \vee \neg Y)}_{\text{tautology}} \wedge (Y \vee X) \equiv X \vee Y$

**Problem 6**

Suppose that the possible values of  $x$  are all people. Let  $Y(x)$  = “ $x$  has green hair,” let  $Z(x)$  = “ $x$  likes pickles” and  $W(x)$  = “ $x$  has a pet frog.” Translate the following statements into words.

- (a)  $(\forall x)Y(x)$ .
- (b)  $(\exists x)Z(x)$ .
- (c)  $(\forall x)[W(x) \wedge Z(x)]$ .
- (d)  $(\exists x)[Y(x) \rightarrow W(x)]$ .
- (e)  $(\forall x)[W(x) \leftrightarrow \neg Z(x)]$ .

**Solution**

**Part (a):** Every person has green hair.

**Part (b):** At least one person likes pickles. // Some people like pickles.

**Part (c):** All people have a pet frog and like pickles.

**Part (d):** For some people, if they have green hair, then they have a pet frog.

**Part (e):** Every person has a pet frog if and only if they do not like pickles.